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### **1. What is Eigenfaces?**

Eigenfaces is a face-recognition technique based on PCA (Principal Component Analysis). It converts face images into a smaller set of important features called eigenfaces and compares them to recognize faces.

### **2. Define PCA in image processing.**

PCA is a dimensionality-reduction technique. It transforms a large number of image features into a smaller number of important components while preserving most of the useful information.

### **3. What is contour-based shape representation?**

Contour-based representation describes an object using its boundary or outline. Examples include chain codes, polygonal approximation, and boundary signatures.

### **4. What is region-based shape representation?**

Region-based representation describes the entire area inside an object's boundary. It uses properties such as area, moments, texture, and pixel distribution to represent the shape.

### **5. How does an active contour (snake) work?**

An active contour, or **snake**, is a curve that moves toward an object's boundary. It uses internal forces to keep the curve smooth and external image forces to attract it toward edges.

### **6. How does a level set method work?**

A level set method represents a contour as a mathematical function rather than directly tracking the curve. The function evolves according to image information until the contour reaches the desired object boundary.

### **7. What is the Hough Transform?**

The Hough Transform is a technique used to detect geometric shapes such as lines and circles in images. For example, in road detection, it can detect straight lane lines from edge pixels.

#### **8. How are Fourier descriptors used in shape analysis?**

Fourier descriptors represent the boundary of an object using Fourier coefficients. They are useful for comparing shapes because they can describe the overall shape while being relatively robust to noise.

#### **9. How are wavelet descriptors used in shape analysis?**

Wavelet descriptors analyse a shape at different scales and frequencies. They can capture both global shape information and small local details.

#### **10. What is a particle filter?**

A particle filter is a tracking algorithm that represents the possible state of an object using many particles. It is useful for tracking objects when there is noise, uncertainty, or partial occlusion.

#### **11. How is foreground–background separation done in surveillance systems?**

The system separates moving objects from the stationary background. Common techniques include background subtraction, frame differencing, and GMM-based modelling. The detected foreground can then be used to identify people or vehicles.

#### **12. How do multi-camera systems track pedestrians?**

Multiple cameras observe the same area from different viewpoints. The system detects pedestrians in each camera and associates the same person across cameras using features such as appearance, position, movement, and camera geometry.

#### **13. What is a Convolutional Neural Network (CNN)?**

A CNN is a deep-learning network mainly used for image and video processing. Convolutional layers automatically learn features such as edges, textures, shapes, and objects from images.

#### **14. How does a Vision Transformer (ViT) work?**

A Vision Transformer divides an image into small patches and converts them into numerical representations called tokens. A Transformer then uses self-attention to learn relationships between different image patches.

#### **15. What is a Generative Adversarial Network (GAN)?**

A GAN consists of two neural networks: a Generator and a Discriminator. The generator creates fake images, while the discriminator tries to distinguish real images from fake ones. Both networks improve through competition.

#### **16. How is an SVM used for image classification?**

An SVM (Support Vector Machine) finds the best decision boundary that separates different image classes. Image features are given as input, and the SVM classifies the image into categories such as car, person, or road.

#### **17. How are Hidden Markov Models (HMM) applied in activity recognition?**

An HMM models activities as a sequence of hidden states. For example, in surveillance, observations such as walking, standing, or running can be used to recognize a person's overall activity.

#### **18. What is a Gaussian Mixture Model (GMM)?**

A GMM represents data as a combination of multiple Gaussian distributions. In computer vision, it can model the background of a scene and identify pixels that significantly differ from the background.

#### **19. How is graph-based learning applied to 3D shape correspondence?**

3D shapes can be represented as graphs, where vertices represent points or regions and edges represent their relationships. Graph-based learning finds corresponding points or regions between two 3D shapes by learning their structural and geometric relationships.

#### **20. How is road and lane detection performed in autonomous driving?**

A typical pipeline is:

Camera → Preprocessing → Edge Detection → Region of Interest → Line/Curve Detection  
→ Lane Tracking → Output

Techniques such as Canny edge detection, Hough Transform, perspective transformation, and deep-learning models can be used to detect lanes. The detected lane information helps the autonomous vehicle understand its driving path.

